SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE CODE: CSA04 **COURSE NAME: Operating Systems**

COURSE OUTCOME COVERED

CO1: *Apply the concepts of Operating System types, structures, and system calls to demonstrate their functions in various computing environments.* *(BL2, BL3)*

QUESTIONS: 5 **Total Marks:100**

Annexure A

Concept Mapping

Code of Conduct:

I Subash G [192572150] certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Subash G

Annexure A

Concept Mapping

Question 1
Show a multi-level concept map integrating:
Operating System Types → OS Structures → System Calls → OS Functions.
Your map must show cross-links (e.g., how OS structure influences system call implementation). **(BL2)**

Question 2
Construct a comprehensive concept map representing:
Process Management, Memory Management, File System, and I/O System.
Include interdependencies and indicate where system calls interact with each component. **(BL3)**

Question 3
Sketch a comparative concept map for:
Batch OS, Time-Sharing OS, Distributed OS, and Real-Time OS.
Highlight differences, similarities, advantages, and limitations using linking phrases. **(BL2)**

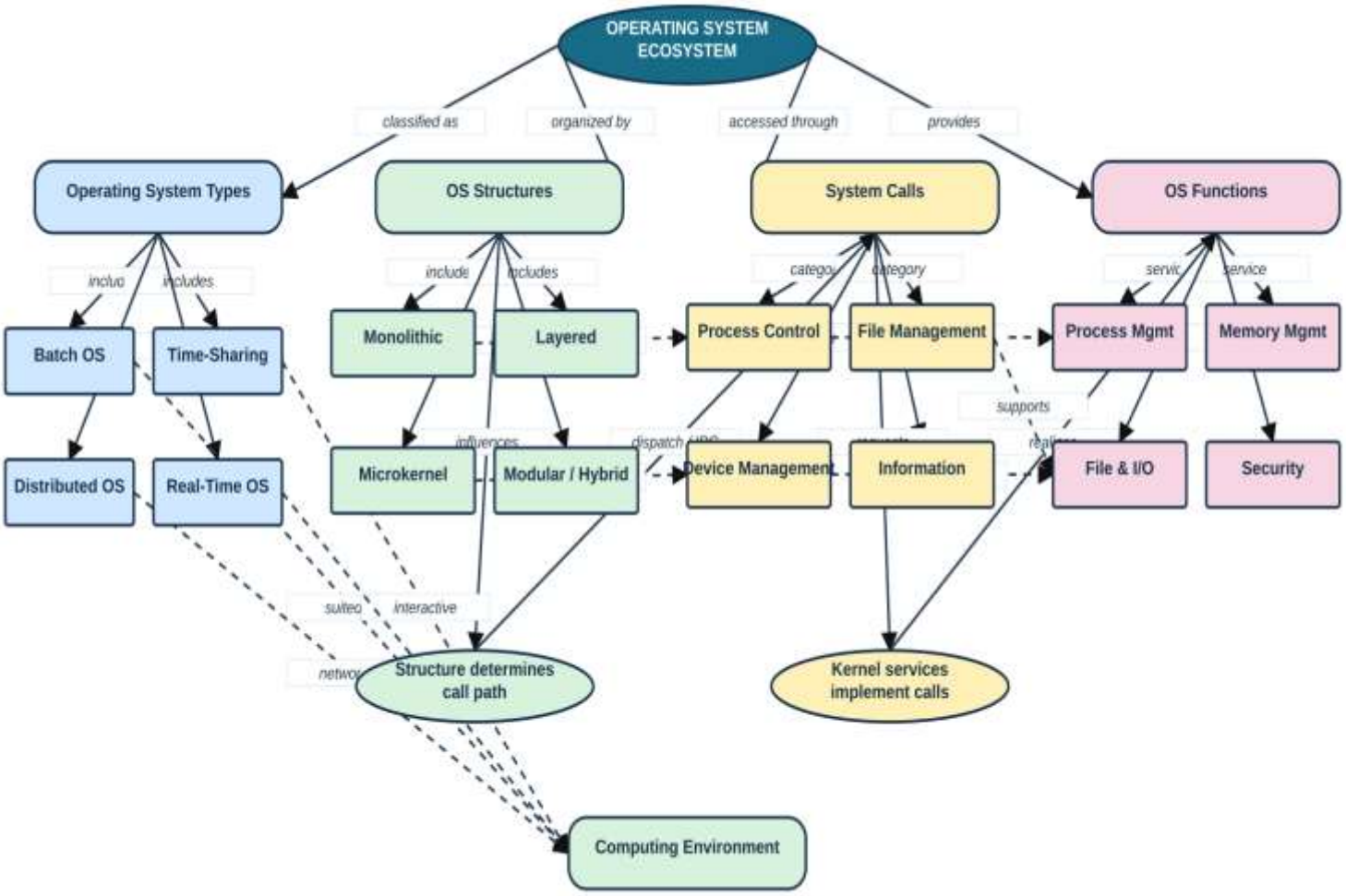
Question 4
Illustrate a dynamic concept map showing the execution flow:
User Program → API → System Call Interface → Kernel → Hardware → Response back to User.
Include feedback loops and control flow paths. **(BL3)**

Question 5
Sketch a hierarchical + functional concept map for system calls:
Process, File, Device, and Information system calls.
Also show how they relate to OS services like scheduling, memory allocation, and device handling. **(BL3)**

RUBRICS FOR CALCULATION

Concept Mapping					
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Question 1 — Operating System Types → OS Structures → System Calls → OS Functions

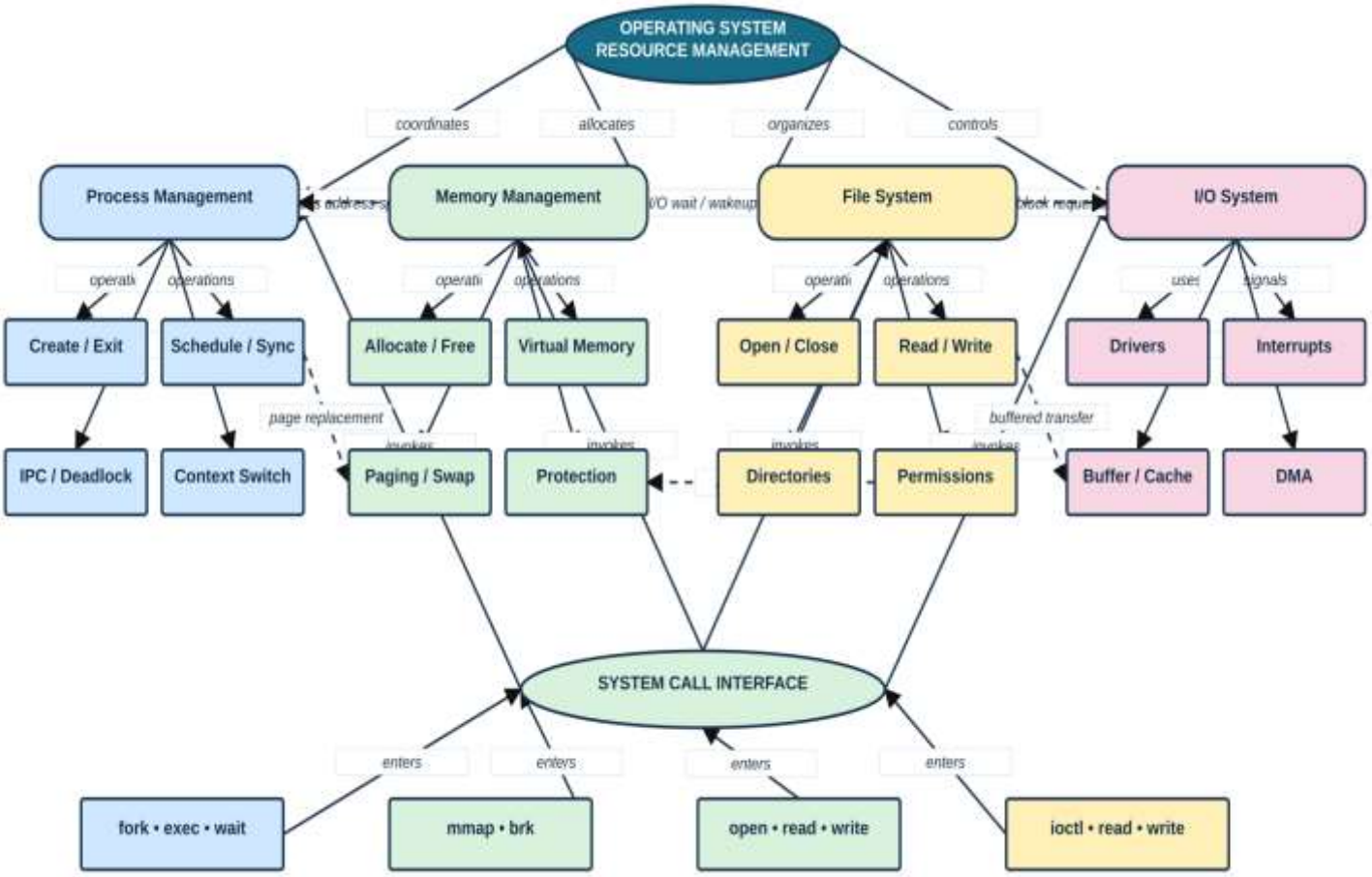


Solid arrows: primary relationship or flow • *Dotted arrows: cross-link, comparison, or feedback*

Important linking phrases:

- OS is classified as → OS Types
- OS is organized by → OS Structures
- OS is accessed through → System Calls
- OS provides → OS Functions
- Structure determines → System Call Path
- Kernel services → Implement System Calls

Question 2 — Process, Memory, File-System and I/O Management

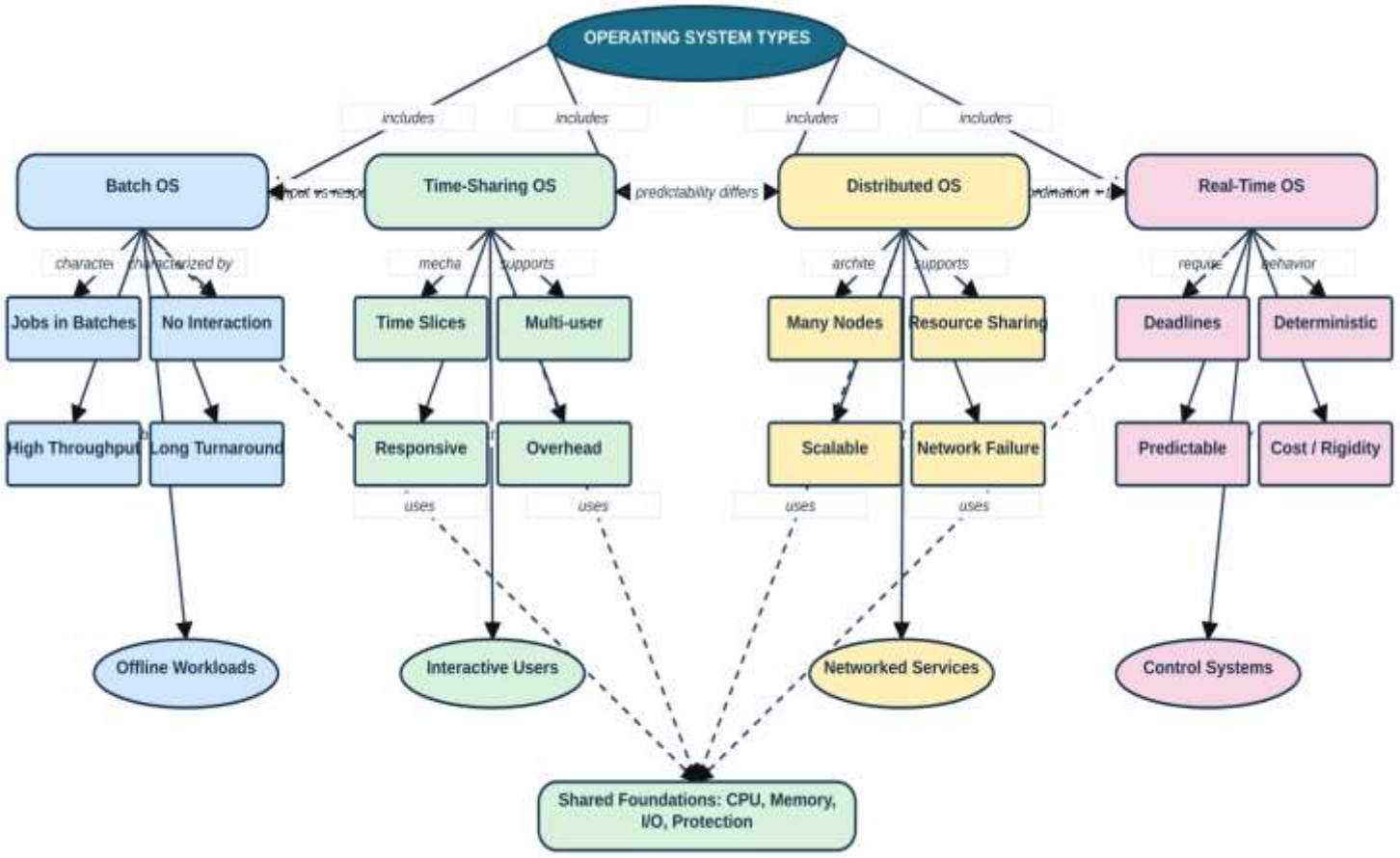


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Question 3 — Comparative Map of Batch, Time-Sharing, Distributed and Real-Time OS

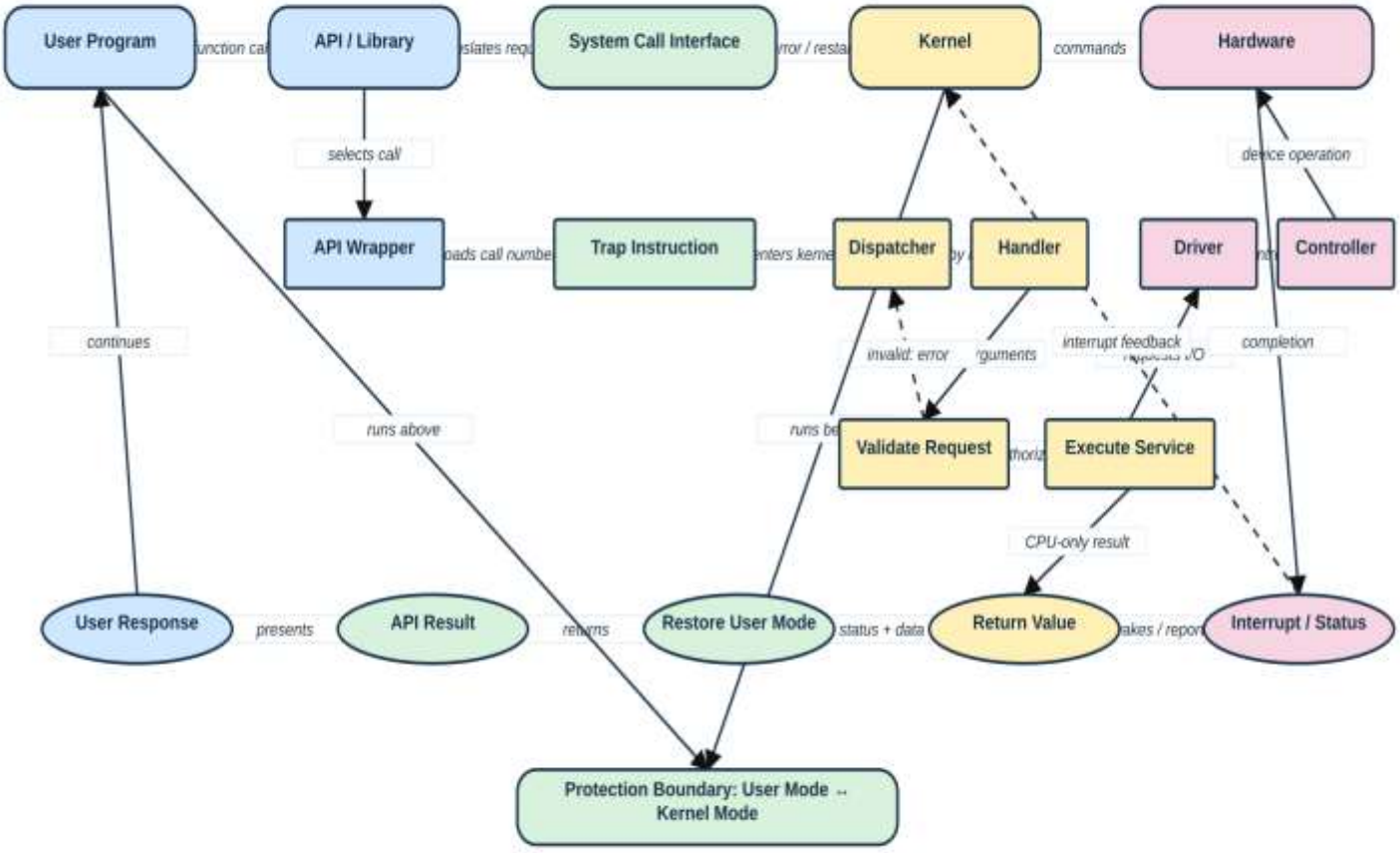


Solid arrows: primary relationship or flow • Dotted arrows: cross-link, comparison, or feedback

Linking phrases:

- Batch OS → characterized by → Jobs in batches
- Time-Sharing OS → supports → Multi-user
- Distributed OS → enables → Resource sharing
- Real-Time OS → requires → Deadlines
- All OS types → share → CPU, Memory, I/O, Protection

Question 4 — System Call Execution Flow and Feedback Loops

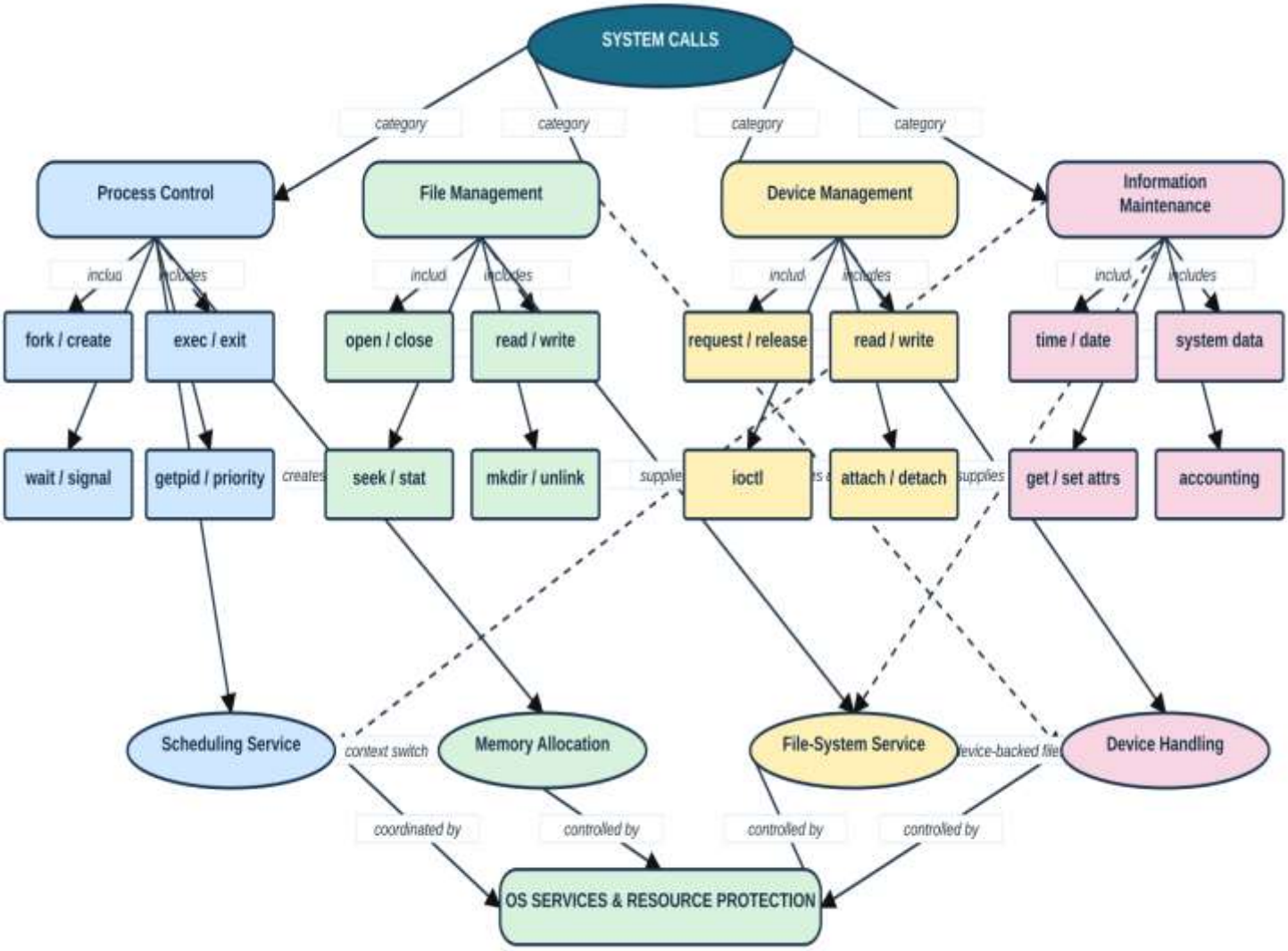


Solid arrows: primary relationship or flow • *Dotted arrows: cross-link, comparison, or feedback*

Important points:

1. System calls provide an interface between user programs and the OS kernel.
2. User programs normally run in user mode, while privileged OS operations are performed in kernel mode.
3. API / Library provides functions that allow applications to request OS services.
4. The System Call Interface identifies the requested system call and transfers control to the kernel.
5. A trap instruction is used to switch from user mode to kernel mode.
6. The kernel validates the request before executing the requested service.
7. The system call handler performs the appropriate kernel operation.
8. Device drivers allow the kernel to communicate with hardware devices.
9. Hardware can generate interrupts to notify the kernel about events or completion.
10. After execution, the kernel returns a value/status to the user program.
11. The system then restores user mode and the application continues execution

Question 5 — Hierarchical and Functional Map of System Calls



Solid arrows: primary relationship or flow • Dotted arrows: cross-link, comparison, or feedback

Important Process:

Process Control: Creates, executes, synchronizes and terminates processes.

File Management: Creates, opens, reads, writes and deletes files.

Device Management: Requests, releases and controls I/O devices.

Information Maintenance: Retrieves and modifies system and process information.:

System Calls → OS Services → Resource Management & Protection